



MASENO UNIVERSITY
UNIVERSITY EXAMINATIONS 2023/2024

**SECOND YEAR SECOND SEMESTER EXAMINATIONS FOR THE
DEGREE OF BACHELOR OF SCIENCE IN AGRONOMY WITH
INFORMATION TECHNOLOGY**

MAIN CAMPUS

AAG 204: PRINCIPLES OF CROP PRODUCTION

Date: 30th April, 2024

Time: 8.30 - 11.30 am

INSTRUCTIONS:

- DO NOT write anywhere on this question paper
- This paper consists of TWO SECTIONS, A and B
- Answer all questions according to the instructions given.



30/4 8:30

350 Gage

AAG 204: PRINCIPLES OF CROP PRODUCTION

Section A: Answer ALL Questions from this section (30 marks)

1. i) The evolution of the Art of Crop Production is as old as civilization, list some of the seven features that have remained unchanged since the dawn of history (2 mks)

iii) Most of the common crops under production today, have been distributed over time from one zone of production to another throughout the world. Name any TWO centers of origin and an example of the crop associated with it (2mks)
2. i) Define the term Cropping system as used in crop production (2mks)

ii) Describe any FOUR practiced cropping systems in Kenya (2 mks)

iii) What are the main factors that determine the crops combination in mixed cropping (2 mks)

iv) A farmer has been producing maize crop in a field for years, producing 2 tonnes per hectare, however, in the last season, he noted a drop production. The farmer decides to intercrop the maize with beans which is known to produce 0.5 tonnes per hectare under proper management; however, the final yields for the two crops were 1.5 tonnes of maize and 0.25 tonnes of beans per hectare when intercropped. What were the relative yields for each of the two crops?

v) Will you advise the farmer to continue with the intercrop or growth the two crops solely? Explain.
3. Differentiate between 'Water requirement of a crop' and 'Field Irrigation requirement of a crop'.
4. i) What are the Essential nutritional elements for plant growth (2 mks)

ii) Factors that determine the amount of fertilizer to be applied in a crop field (2 mks)

iii) A farmer wanted to apply 120kg nitrogen, 60kg phosphorous and 60kg of potassium per hectare in a tobacco field from a compound fertilizer of N15-P15-15 packed in 50kg bags. How many bags of this fertilizer will be required to meet the supply in two hectares of tobacco crop? (3 mks)
5. i) What are the purposes of tillage (2 mks)
ii) How does the depth of tillage affect crop establishment and growth (1 mk)

Section B: Answer Question 1 and TWO others from this Section (40 marks)

1. Comment on the following as applied in crop management practices (15 mks)
 - i) Methods of controlling Diseases and Pests
 - ii) Integrated Weed management
 - iii) Agronomic benefits of using Mulches and green manures
2. Discuss the main agronomic measures to control Soil Erosion (25 mks)
3. i) Outline the factors affecting Irrigation and water Efficiency of a crop (10 mks)
ii) State how the above factors can be eliminated (15 mks)
4. Giving reasons for preferences, discuss the different types of tillage applied in crop production (25 mks)
5. With examples, discuss the types, sources and forms of plant nutrients (25 marks)